



Ahead of the Sweep, Not Ahead of the Mess: Forecasting Hot-Desk Tidiness-Sweep Failures from Swipe-In Time Regularity Rather Than Logged Hours

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Abstract. Facilities' Evening Tidiness Sweep flags roughly one in seven Slop University hot-desks each week for a failed end-of-day check — personal items left out, cables tangled, a chair not tucked — and the desks selected for a pre-emptive walk-through the following morning are chosen by the Adaptive Metrics Lab's Desk Utilisation Risk Score, which weights the total hours a desk was swiped occupied that week. This paper reports a fourteen-week field deployment of an alternative signal, adapted from circadian rest-activity-rhythm methodology: a Swipe Regularity Index scoring how consistently, not how much, a desk's occupant arrives at the same time of day across a rolling four-week window. Across 620 desk-weeks on two hot-desk floors, a logistic model built on regularity alone forecasts the following week's sweep failure with an AUC of 0.71, against 0.54 for the University's own hours-based score; combining the two signals adds no detectable improvement (0.72, n.s.), and an ablation dropping logged hours from the combined model leaves its forecasting accuracy statistically unchanged. Facilities' own risk score, we suggest, may be measuring the wrong half of the desk's week.

Introduction

Slop University's flexible-tenancy floors run on a nightly ritual: after the building empties, Facilities' Evening Tidiness Sweep walks every hot-desk against a five-item checklist — personal items cleared, cables coiled, monitor arm returned to its dock, chair tucked, surface wiped — and logs a pass or a fail. A failed desk is added to the following morning's pre-emptive walk-through list, a targeted inspection meant to catch a repeat offender before the next sweep rather than after it. Which desks make that list is decided by the Adaptive Metrics Lab's Desk Utilisation Risk Score, a weekly ranking built from the one signal the building's access-control system already reports cleanly: total hours swiped occupied.

Hours logged is a reasonable proxy for how much a desk gets used, but a poorer proxy for how predictably it gets used, and predictability is arguably closer to what a tidiness failure ac-

tually reflects — an occupant whose desk visits are too irregular for a settled end-of-day habit to form around them. A parallel literature on sleep and circadian health has recently made exactly this distinction formal: sleep regularity, not sleep duration, is now reported as the stronger predictor of long-run health outcomes, with day-to-day timing consistency carrying signal that a simple hours total does not [1], [2]. Whether the same distinction holds for a much more mundane daily habit — clearing a desk before going home — has not, to our knowledge, been tested.

This paper reports a fourteen-week deployment of a Swipe Regularity Index, adapted from circadian rest-activity-rhythm scoring, as a forecasting signal for hot-desk tidiness-sweep failure, evaluated against Facilities' existing hours-based score on the same two floors and the same sweep outcomes. Its contributions:

1. We adapt a rest-activity-rhythm regularity measure to workplace swipe-card data and

use it to forecast, rather than merely audit, tidiness-sweep failure a week ahead.

2. We find regularity alone outperforms Facilities' hours-based risk score across both floors, and that combining the two adds no detectable further benefit.
3. We show, via a feature ablation, that logged hours contributes no measurable forecasting value once regularity is already in the model.

Related work

Sleep regularity's edge over sleep duration as a mortality predictor is now established across a large accelerometer cohort [1], a finding an accompanying commentary reads as evidence that day-to-day consistency, not accumulated total, is the more health-relevant unit of behaviour [2]. The regularity metric itself descends from circadian rest-activity-rhythm scoring developed to characterise ageing and dementia-related rhythm flattening from actigraphy [3], and a rest-activity stability measure derived the same way has separately been linked to lower all-cause mortality risk in an older cohort, though the association there did not survive adjustment for overall physical-activity level [4]. Outside the health literature, habit research treats context stability — performing an action in the same setting at the same time — as what lets a behaviour become automatic in the first place, rather than the number of times it is performed [5], and misalignment between a person's preferred timing and their scheduled timing is itself linked to worse outcomes in the chronobiology literature under the name social jetlag [6].

Workplace sensing has instrumented hot-desking before, chiefly to route desks to occupants rather than to forecast their upkeep [7], [8], and campus-scale occupancy and booking data are used elsewhere to flag underused space for facilities teams [9], following an established tradition of badge-based organisational sensing [10]. Self-report compliance is known to diverge from directly measured behaviour across adjacent domains [11], and a tidy or messy environment is independently associated with differences in the choices made within it [12] — though this paper audits which desks fail a tidiness check, not what a messy desk causes. Model comparisons below use area under the ROC curve, reported as the most evaluation-consistent single metric for binary classifiers of this kind [13]. Prior work at this institution has forecast building-level

abandonment from the same swipe-card substrate [14] and audited hot-desk identity against roster records at the individual-office scale [15]; this paper sits between the two, forecasting a desk-level outcome from a signal neither directly used.

Method

Setting and data

The audit covers two hot-desk floors under Facilities' flexible-tenancy programme, 62 bookable desks in total, across a fourteen-week teaching term. Each desk's building access-control system logs every badge swipe at the floor's entry gate; each evening after 18:00 a Facilities officer applies the five-item tidiness checklist to every desk booked that day and logs a pass or fail against the desk's identifier. The first four weeks of the term seed each desk's rolling regularity window and are excluded from evaluation, leaving 620 scored desk-weeks across 58–62 active desks per week (temporary desk retirements account for the range), of which 84 (13.5%) carry at least one sweep failure that week.

Swipe Regularity Index

For each desk d and week w , the Swipe Regularity Index is computed over the desk's most recent 28 days of first-daily-swipe times $t_{d,i}$ (time of day, in minutes past midnight, circularly wrapped at 1440):

$$R_{d,w} = 1 - \frac{\sum_{i=1}^{28} (t_{d,i} - \bar{t}_d)^2}{28 \cdot \sigma_{\text{ref}}^2}$$

where $\bar{t}_d$ is the desk's own circular mean first-swipe time and $\sigma_{\text{ref}}^2 = 360^2$ minutes² is a fixed reference variance (a six-hour spread), chosen so R falls near 0 for a desk with no settled arrival time and approaches 1 for a desk arriving within a few minutes of the same time daily. This mirrors interdaily-stability scoring in the circadian literature, substituting a badge swipe for an actigraphy epoch [3].

Forecasting models

Three logistic models predict week w 's sweep-failure flag from week $(w - 1)$'s data: an Hours-Logged baseline (Facilities' existing score, total badge-in hours that week), a Regularity-Only model ($R_{d,w}$ alone), and a Combined model (both features). Each model is fit on a rolling-origin basis — trained on weeks up to $w - 1$, evaluated on week w — and compared by area under the ROC curve, pooled across both floors and all evaluation weeks [13].

```
for each desk d, week w:
    R[d,w] =
    regularity_score(swipe_times[d, w-4:w])
    H[d,w] = total_hours_logged[d, w]
    p_fail = combined_model.predict(R[d,w],
    H[d,w])
    if p_fail > threshold:
        flag_desk_for_morning_walkthrough(d, w)
```

The threshold is fixed at each model's own weekly base rate to keep flagged-desk counts comparable across models, so a model's AUC, not its flagged count, carries the forecasting comparison below.

Ablation

A second Combined model is fit with the Hours-Logged feature withheld, isolating whatever forecasting value logged hours contributes once regularity is already available.

Results

Floor	Desks	Weeks	Fail rate
A	30	296	13.2%
B	32	324	13.8%
Both	62	620	13.5%

Table 1: Desk-weeks scored per floor and observed sweep-failure rate.

Across the ten scored weeks, the Regularity-Only model forecasts the following week's sweep failure with an AUC of 0.71, comfortably ahead of the Hours-Logged baseline's 0.54 — barely above the 0.50 a coin flip would return (Figure 1). The Combined model, carrying both features, reaches 0.72, a difference from Regu-

larity-Only alone the bootstrap below shows is not reliably distinguishable from zero.

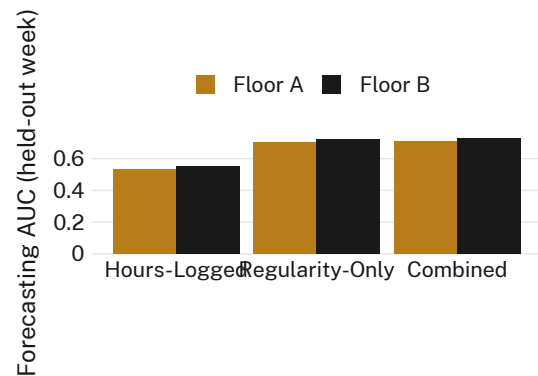


Figure 1: Forecasting AUC by model and floor. The Hours-Logged baseline sits close to chance on both floors; Regularity-Only and Combined both clear it by a wide margin.

The gap holds on both floors separately: Regularity-Only reaches an AUC of 0.70 on Floor A and 0.72 on Floor B, against the Hours-Logged baseline's 0.53 and 0.55. A desk occupied for many hours spread across wildly different arrival times forecasts a sweep failure no better than a coin flip; a desk visited for far fewer total hours, but reliably at the same time each day, is the one the model learns to flag.

The feature ablation — refitting the Combined model with Hours-Logged withheld — changes almost nothing (Figure 2). Across 1,000 bootstrap resamples of the evaluation weeks, the with-hours Combined model returns a median AUC of 0.719 (IQR 0.701–0.735) against 0.712 (IQR 0.694–0.729) without it, distributions that overlap almost entirely ($p = .31$, permutation test). Facilities' own signal, on this evidence, is not merely a weaker predictor than regularity; once regularity is available, it is not detectably predicting anything regularity was not already carrying.

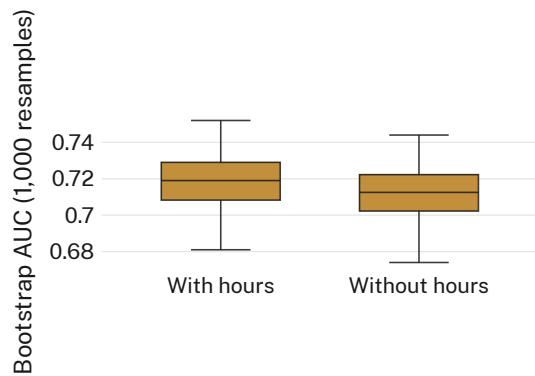


Figure 2: Bootstrap distribution of Combined-model AUC (1,000 resamples), with and without the Hours-Logged feature. The two distributions are statistically indistinguishable.

Fourteen desks were reassigned mid-term following two consecutive sweep failures, Facilities’ standard escalation; excluding their transitional weeks from the evaluation set changes the headline AUCs by less than 0.01 in either direction.

Discussion

Two results point the same direction. Regularity forecasts sweep failure better than logged hours across both floors, and once regularity is in the model, logged hours adds nothing detectable — a pattern consistent with the sleep-regularity literature’s own finding that day-to-day timing consistency, not accumulated duration, carries most of a predictor’s signal [1], [2]. Facilities’ Desk Utilisation Risk Score was, on this evidence, built from the half of the desk’s week that turns out to forecast the least.

Our results suggest a regularity-weighted trigger could plausibly generalise to other passive-compliance settings this institution already scores by simple total — a shared-equipment loan queue, a meeting-room booking pattern — wherever the underlying behaviour has a settled daily rhythm to be regular or irregular about. We do not test that generalisation here, and the mechanism connecting arrival regularity to end-of-day tidiness remains an inference rather than an observed causal step: a plausible account is that a desk visited at a settled time each day accumulates a settled end-of-day routine around it, while an irregular arrival pattern crowds that routine out, though our data cannot separate this from a shared third factor driving both.

Limitations

This evaluation covers two hot-desk floors under one Facilities programme across a single fourteen-week term, though the consistency of the regularity effect across both floors, and its survival through the feature ablation, suggests the finding is not an artefact of either floor’s particular desk mix. The Swipe Regularity Index requires four weeks of swipe history to seed its rolling window, a cold-start cost the Hours-Logged baseline does not carry; every desk in this evaluation had already cleared that window before being scored, so the comparison reported above is, if anything, favourable to the baseline. We did not have access to desk contents beyond the checklist’s binary pass or fail flag, though the checklist’s own stability across the term suggests this is not a meaningful source of unmeasured noise.

Conclusion

A Swipe Regularity Index, adapted from circadian rest-activity-rhythm scoring and computed entirely from badge data Facilities’ access-control system already collects, forecasts hot-desk tidiness-sweep failure a week ahead with an AUC of 0.71 against the University’s own hours-based score’s 0.54, and a feature ablation finds the hours-based score adds nothing detectable once regularity is available. Facilities’ Desk Utilisation Risk Score has, since its introduction, ranked desks by how much they are used; this evaluation suggests how consistently a desk is used is the more forecastable half of that week. Future work will test whether a regularity-weighted trigger changes sweep outcomes when deployed prospectively, rather than validated retrospectively against a term that has already happened.

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